

## Repair Manuals

Timing chain: removal/installation

### Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

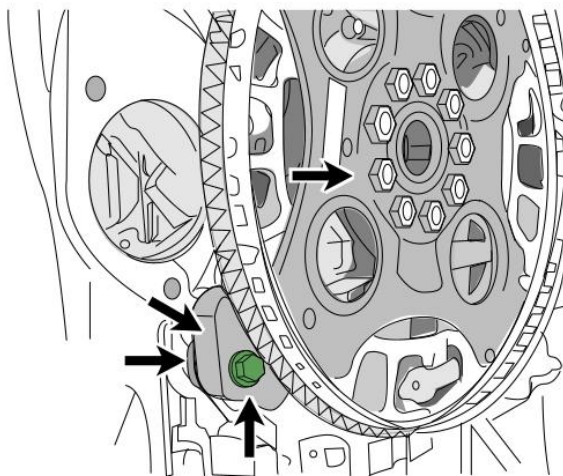
Always use new gaskets

### Removal

Remove the engine

Vehicles with automatic transmission:

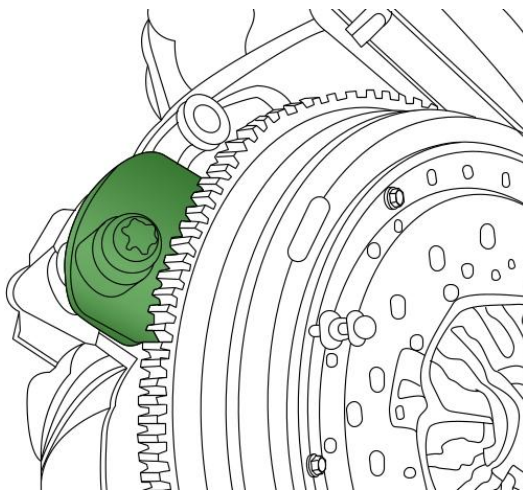
Fit the locking tool (11 9 260)



Remove the flexplate

Vehicles with manual gearbox:

Fit the flywheel locking tool (11 9 810, 11 9 811)



Remove the flywheel

All vehicles:

Remove the cylinder head

Remove the oil sump

Remove the oil pump

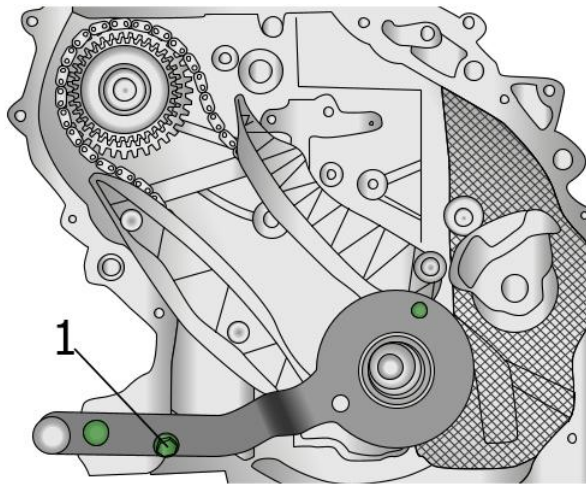
Vehicles with automatic transmission:

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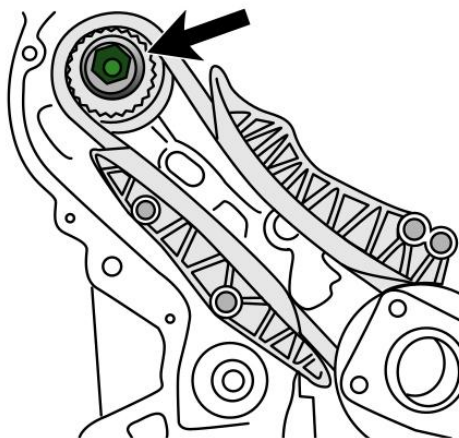
Remove the oil cooler  
Remove the oil cooler bracket

All vehicles:

Remove the crankshaft position sensor ring  
Remove the timing chain cover  
Remove the oil pump chain  
Remove the primary timing chain  
Remove the upper timing chain guide  
Fit the crankshaft locking tool (2 357 904)  
Fit the special tool at the point shown  
Fit the bolt (1 )

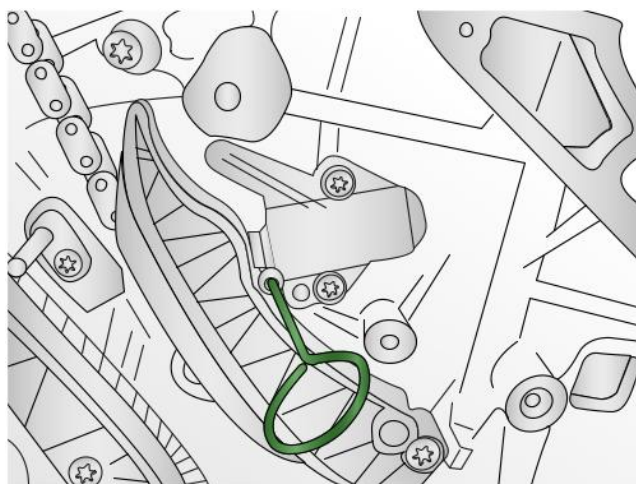


Loosen the high-pressure pump gearwheel bolt  
Fit the special tool: (11 8 740)  
Detach the high-pressure pump gearwheel



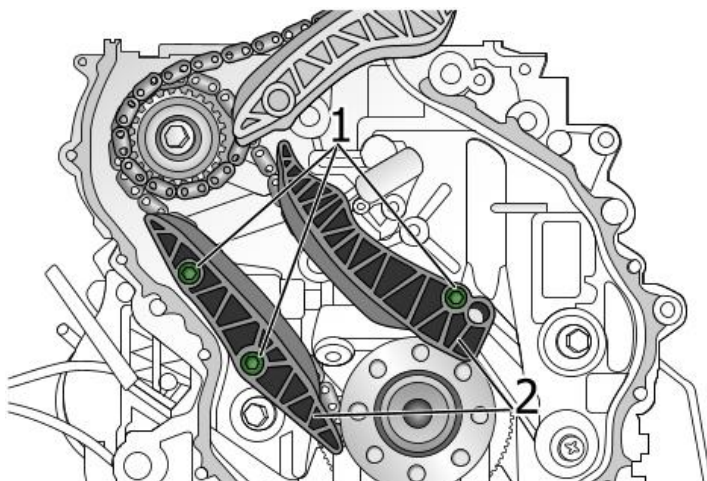
Lock the chain tensioner with a locking pin (11 4 120)

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Remove the secondary chain tensioner

Remove the timing chain guides (1 - 2)



Remove the high-pressure pump gearwheel

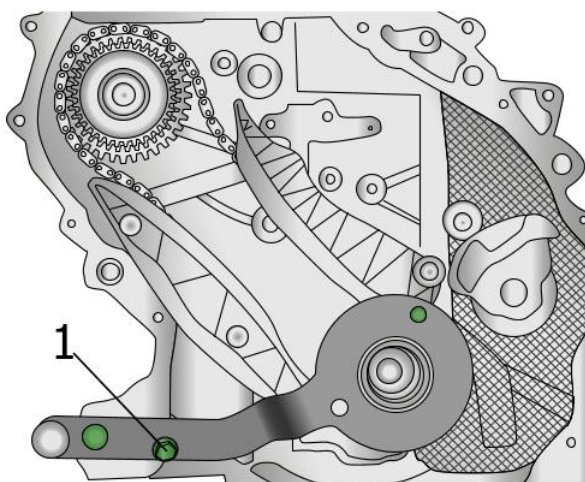
Remove the crankshaft locking tool (2 357 904)

Remove the secondary timing chain

### Installation

Refit the secondary timing chain

Fit the crankshaft locking tool

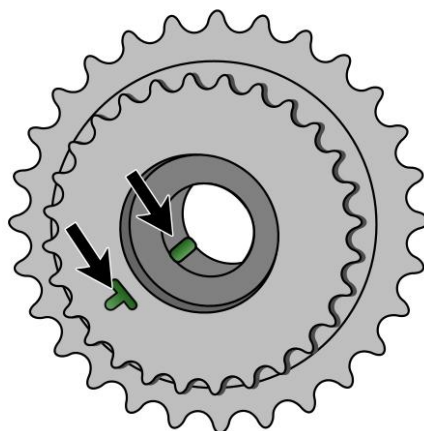
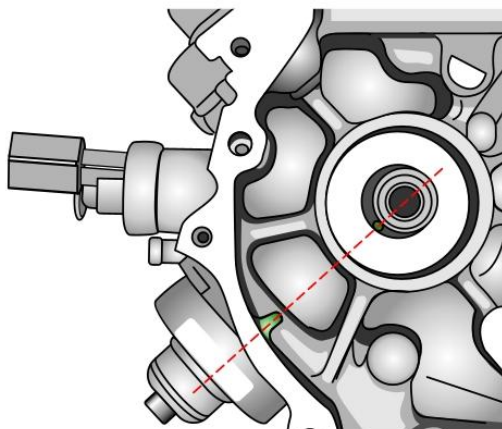


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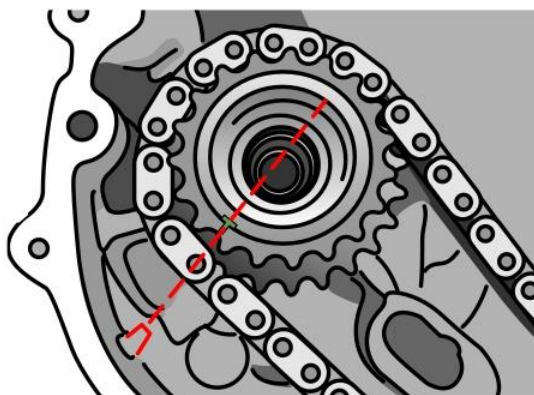
Fit the special tool at the point shown

Fit the bolt (1)

Check the timing marks



Refit the high-pressure pump gearwheel



Timing chain cover:

Check the dowel pin(s) for damage

Apply sealant

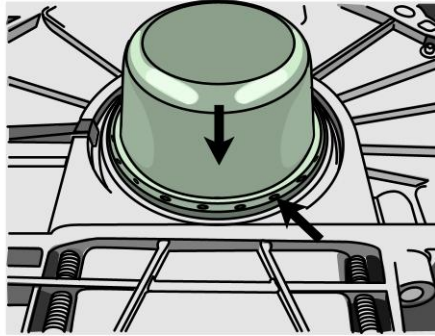
## Repair Manuals

Refit the timing cover

Crankshaft oil seal:

Use the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Fit the special tool: (11 8 815)



Note: Do not apply grease

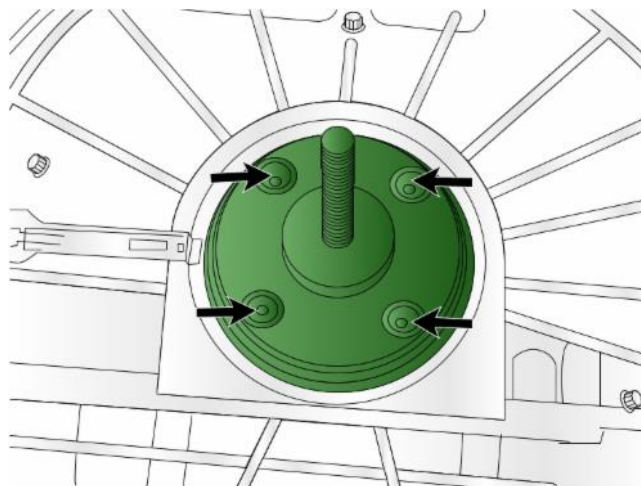
Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal

Remove the special tool (11 8 815)

Fit the special tool: (11 8 811)

Tighten the bolts (11 8 812)

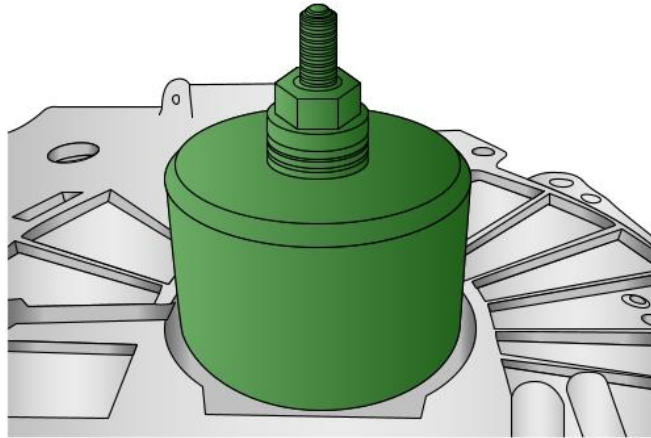


Fit the special tool: (11 8 813)

Tighten the nut (11 8 814)

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Note: Until the crankshaft seal is level with the timing cover

Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Further refitting is in the reverse order of removal

### **Torque settings**

Oil sump

Sump 24 (Nm)

Oil pump

Oil pump 22 (Nm)

Oil pump gearwheel

Oil pump gearwheel (Renew the bolt (left-hand thread))

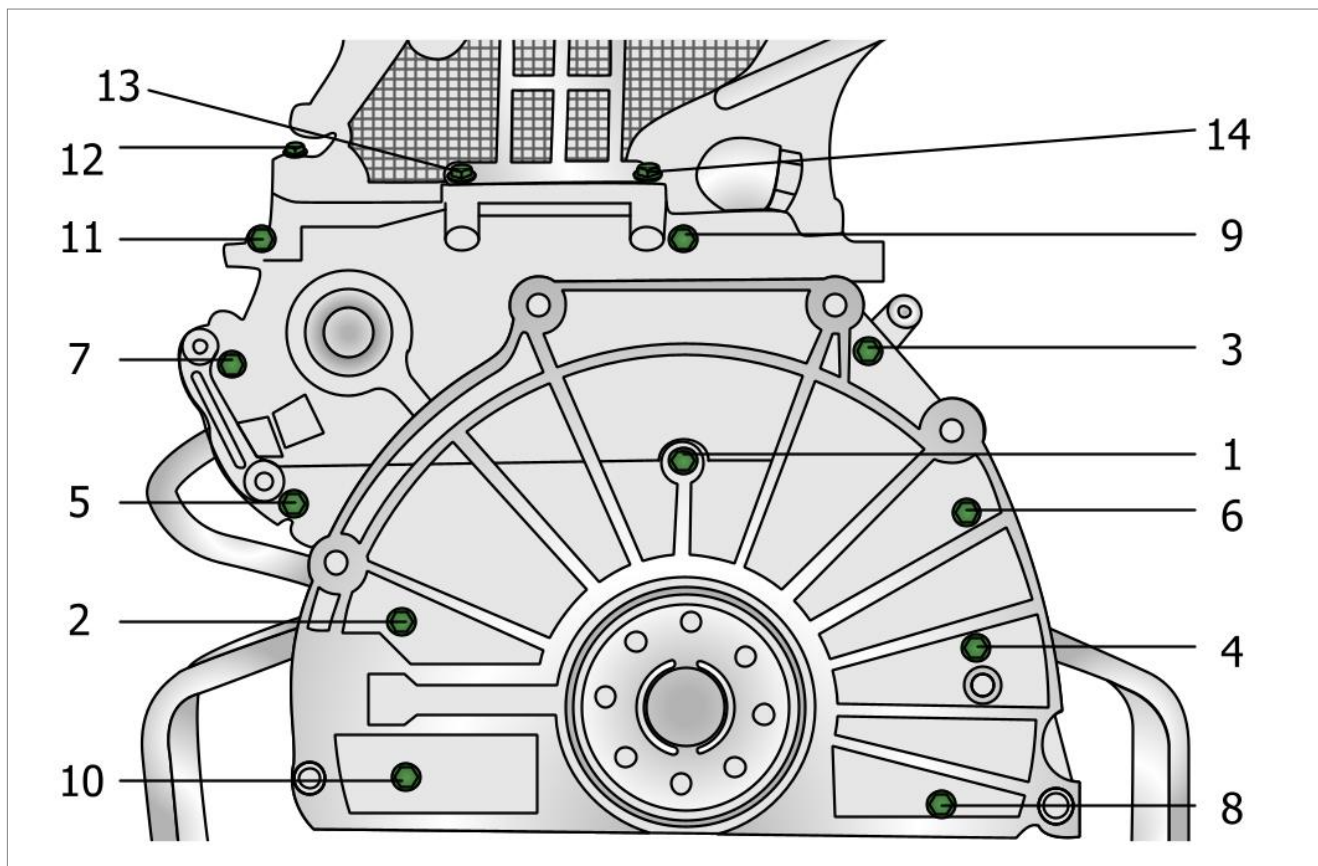
Stage 1 5 (Nm)

Stage 2 90 (°)

Timing cover

Timing cover (Renew the bolts)

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Timing cover (Bolts 1 - 11)

Stage 1 8 (Nm)

Stage 2 90 (°)

Timing cover (Bolts 12 - 14) 19 (Nm)

Timing chain tensioner

Timing chain tensioner (Upper) 70 (Nm)

Timing chain tensioner (Lower) 10 (Nm)

Timing chain guide

Timing chain guide (Renew the bolts) 20 (Nm)

High-pressure pump gearwheel

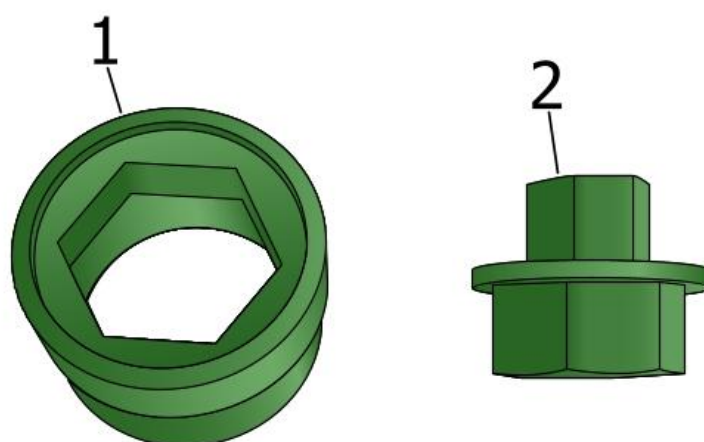
High-pressure pump gearwheel 65 (Nm)

Flywheel

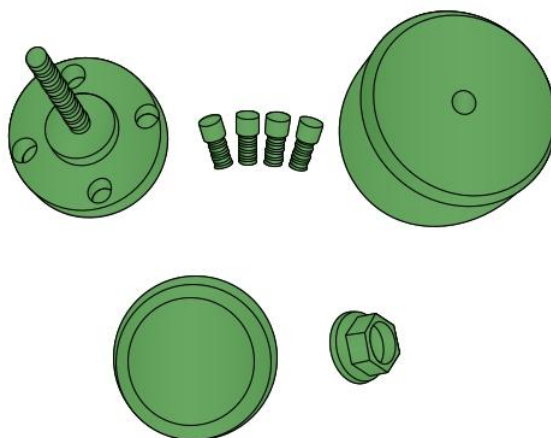
Flywheel (Renew the bolts) 120 (Nm)

### Special tools

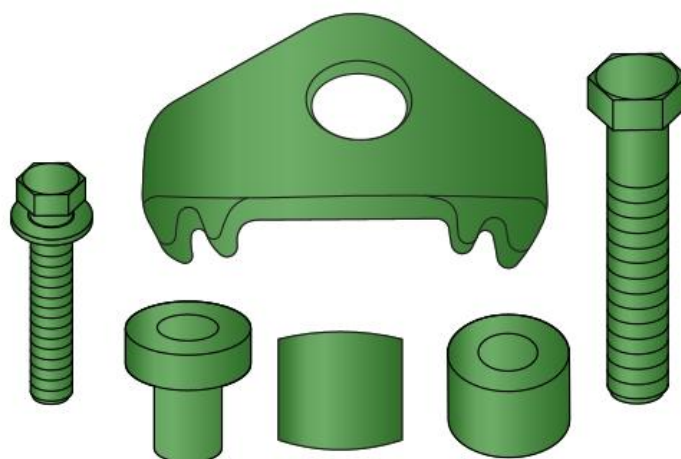
Extractor: (11 8 740)

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Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

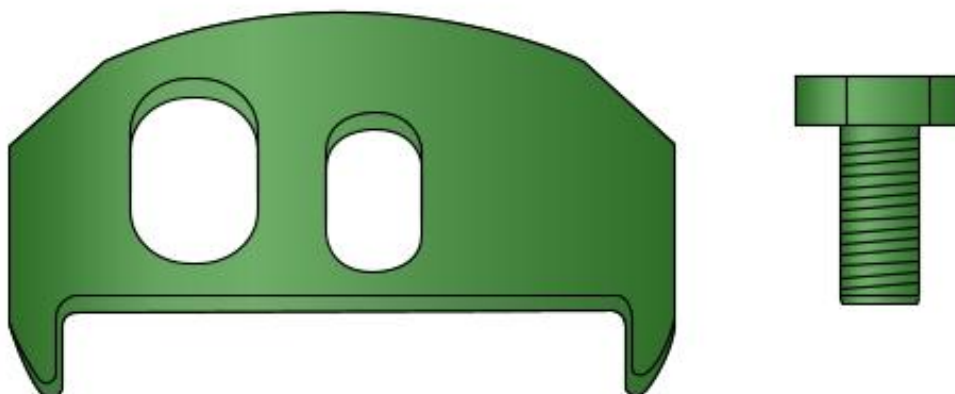


Flywheel locking tool: (11 9 260)

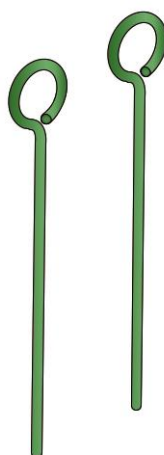


Flywheel locking tool: (11 9 811)

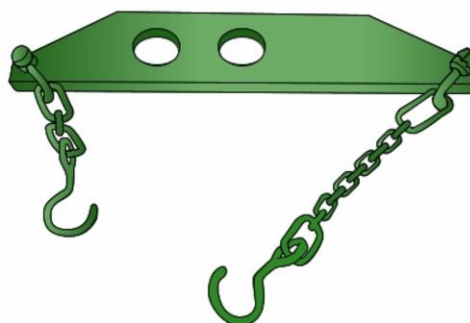
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Locking pin: (11 4 120)



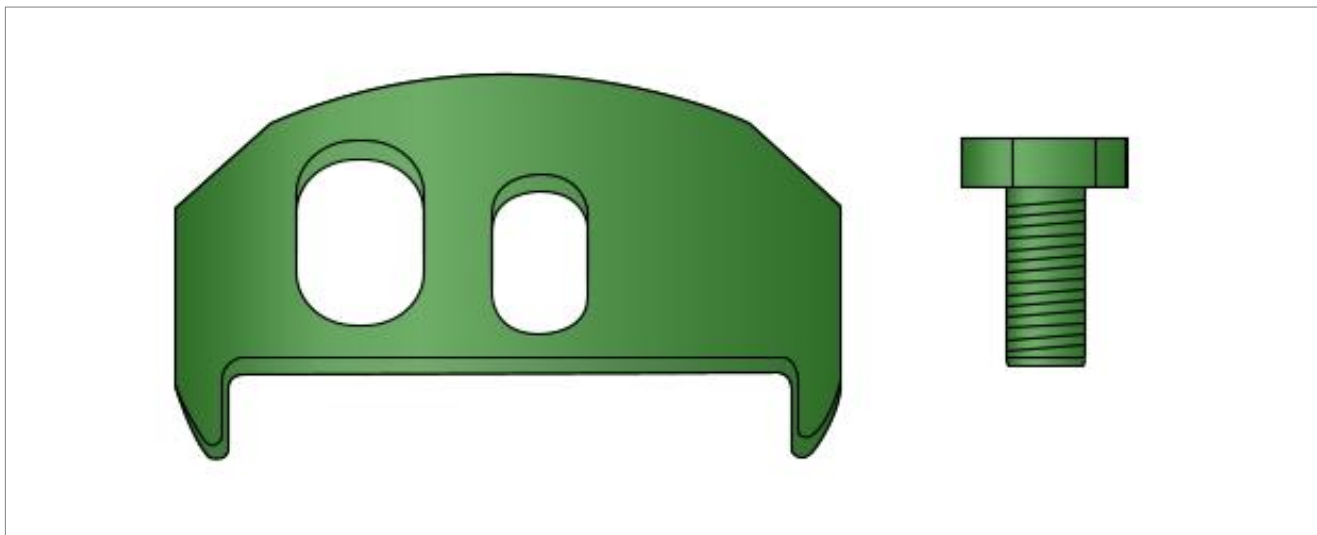
Engine support tool: (11 0 000)



Flywheel locking tool: (11 9 810)

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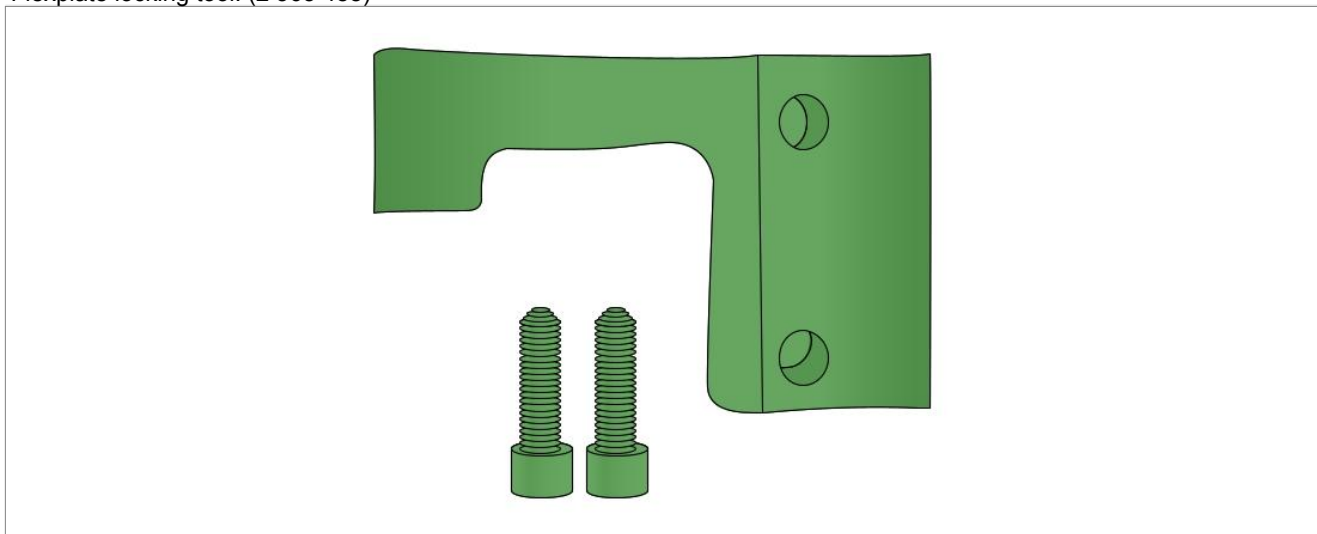
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Crankshaft locking tool: (2 357 904)



Flexplate locking tool: (2 365 488)



Flexplate locking tool: (2 288 380)

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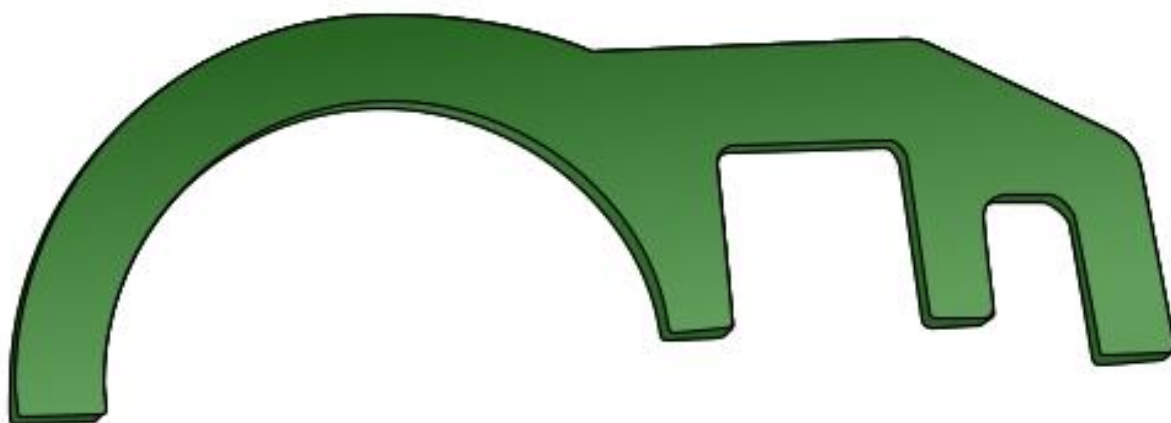
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Socket: (11 6 480)



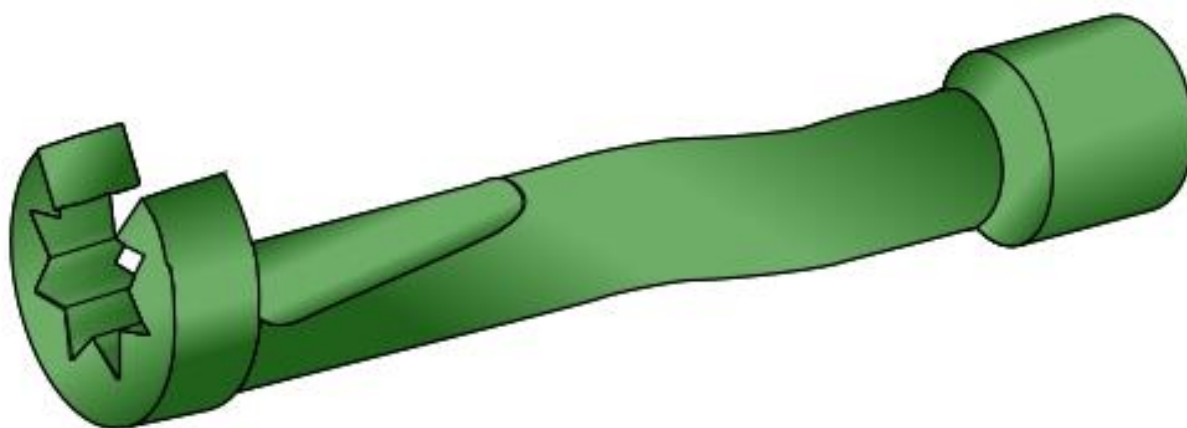
Camshaft blocking tool: (11 8 760)



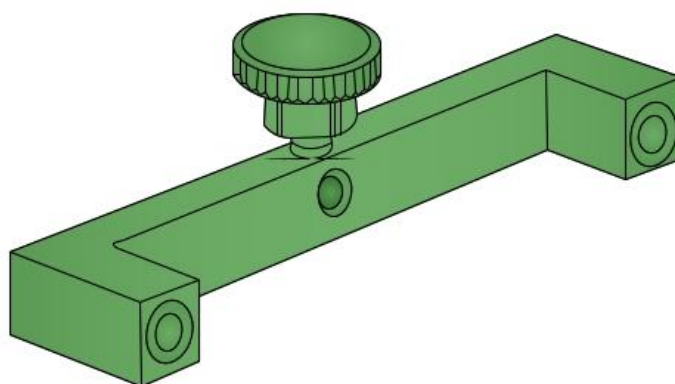
Socket wrench: (37 1 152)

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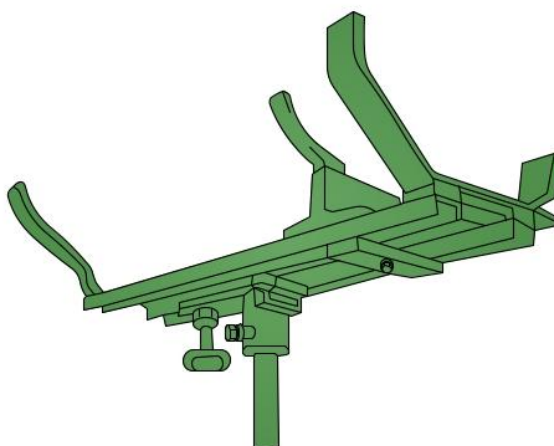
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Dial gauge holder: (00 2 530)



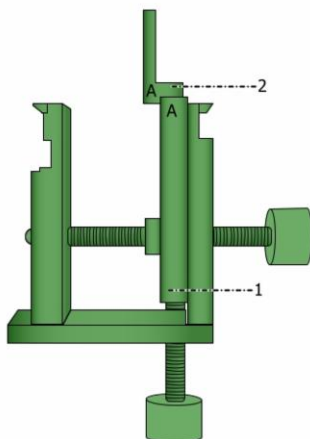
Gearbox support: (23 4 050, 00 2 030)



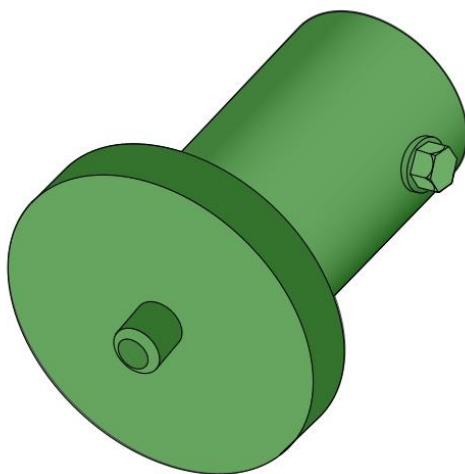
Bearing bracket: (24 4 160, 24 4 161, 24 4 164)

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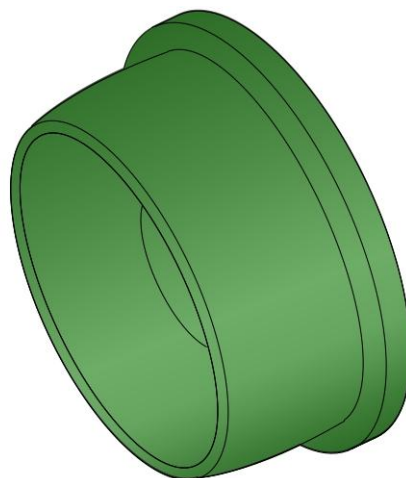
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Support (31 2 220)

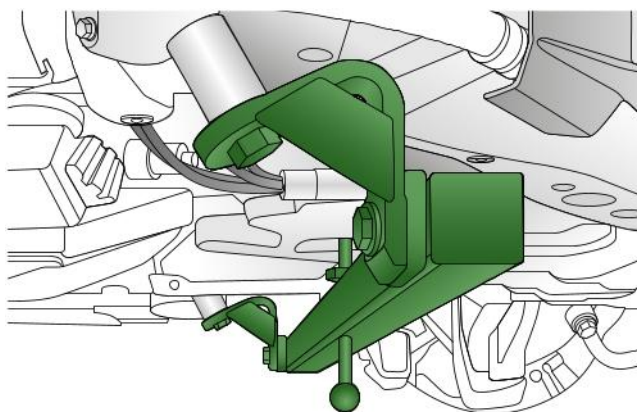


Plug (24 2 390)

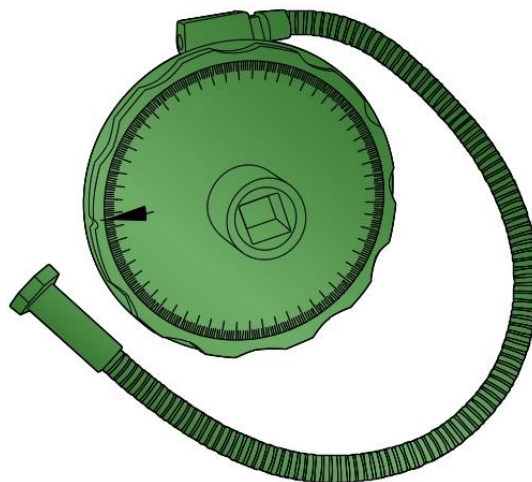


Gearbox support tool: (11 8 910)

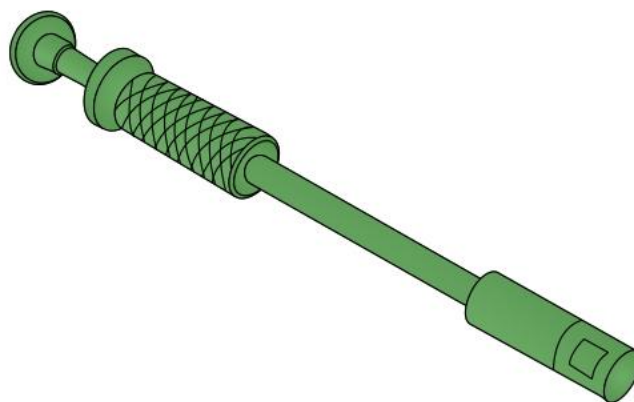
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Torque wrench: (00 9 120)



Injector removal tool: (13 5 250)



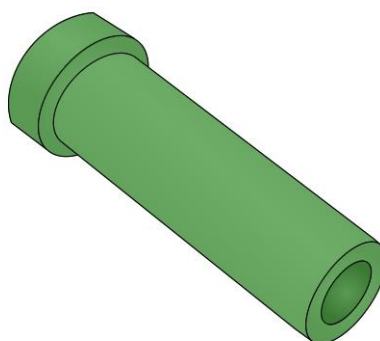
Injector removal tool: (13 5 252)

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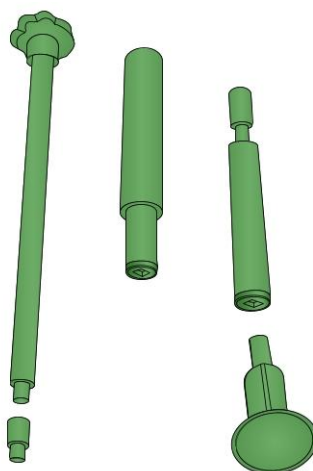
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Injector removal tool: (13 5 253)



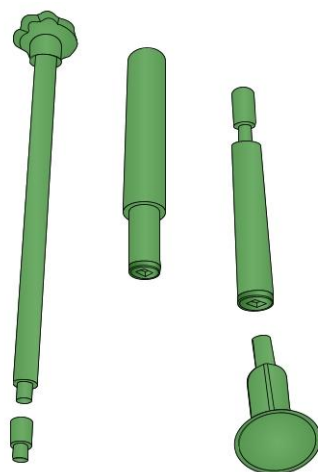
Injector cleaner: (13 5 215)



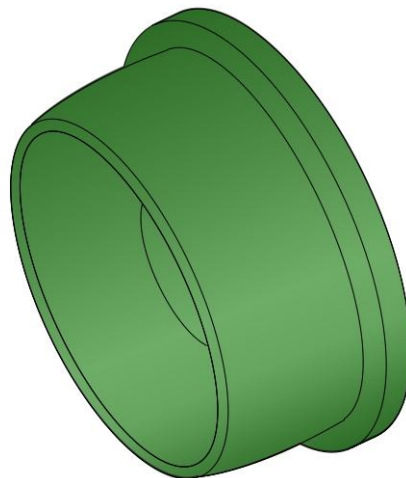
Injector cleaner: (13 5 210)

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Plug (24 3 390)



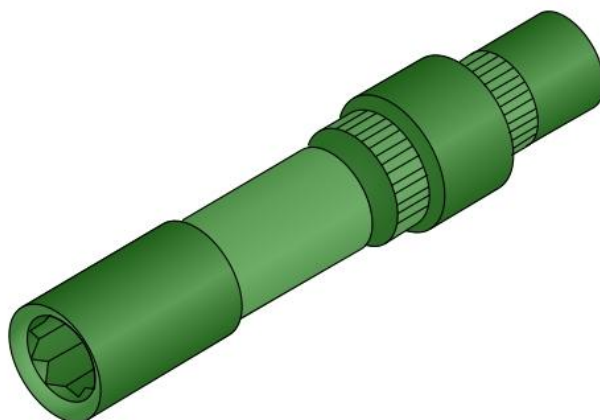
Impact extractor: (13 6 100)



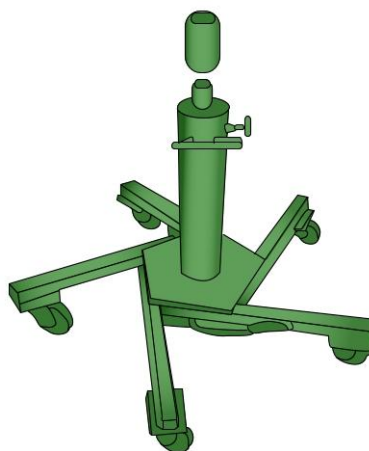
Socket: (11 6 050)

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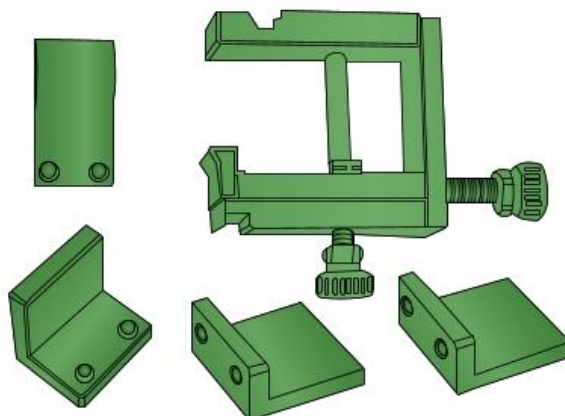
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Transmission jack: (00 2 030)



Locking tool: (24 4 161, 24 4 165)



Gearbox support tool: (23 4 050)

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